# Chapter 2: The Physical Layer

1 What is the basic goal of the physical layer?
物理层的基本目标是什么？
2 How to define the physical layer protocols?
如何定义物理层的协议
3 What is the theoretical basis for the physical layer?
物理层的理论基础是什么？
*4 What are the commonly used transmission media?
常用的传输介质有哪些？
*5 What are the common physical layer interface standards?

- 常见的物理层接口标准有哪些？

# Keypoints and Difficulties

## Keypoints:

- Nyquist's Law and Shannon-Hartley theorem
- 奈奎斯特定律和香农-哈特利定理
- Symbol code types: NRZ, Manchester encoding, Differential Manchester encoding

符号编码类型 NRZ、曼彻斯特编码、差分曼彻斯特编码

## Difficulties:

- Channel transmission rate calculation
- 信道传输速率计算

# 1 The Goals

- To ensure the correctness of the transmitted signals "0" and "1" and the consistency of transmission and reception;确保传输信号 "0 "和 "1 "的正确性以及传输和接收的一致性
- Bit – transmission mode, speed, duration, and signal distortion;比特 - 传输模式、速度、持续时间和信号失真；
- Interface design: the number of pins, specifications, functions, etc.;接口设计：引脚数量、规格、功能等；
- Signal transmission procedures: how to arrange the transmission process and the order of events;
- 信号传输程序：如何安排传输过程和事件顺序；

# Four Important Characteristics

- Mechanical Characteristics机械特性
  - Specifies the size of the connector used in the physical connection, the number and arrangement of pins.
- 指定物理连接中使用的连接器尺寸、针脚数量和排列。
- Electrical Characteristics电气特性
  - Specifies the transmission mode, voltage level, coding, impedance matching, transmission rate and distance limit when transmitting the binary bit.指定传输二进制位时的传输模式、电压电平、编码、阻抗匹配、传输速率和距离限制。

# Four Important Characteristics

- Functional Characteristics功能特点
  - Define the function of each physical line, indicating the means of a special voltage appears on a line定义每条物理线路的功能，说明线路上出现特殊电压的手段
  - Line functions are divided into four categories: Data, Control, Timing, Power Supply线路功能分为四类：数据、控制、计时、供电
- Procedural Characteristics程序特点
  - Define the working rules and timing relationships for each physical line定义每条物理线路的工作规则和时序关系
  - Signal transmission: simplex, half duplex, full duplex
  - 信号传输：单工、半双工、全双工



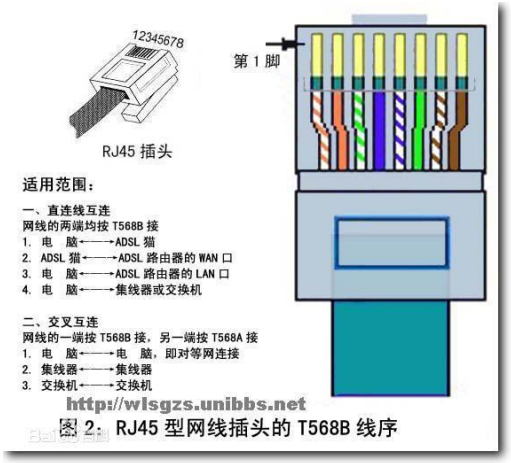
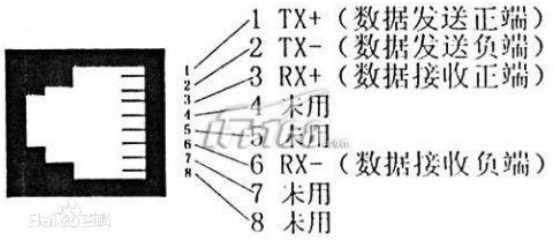
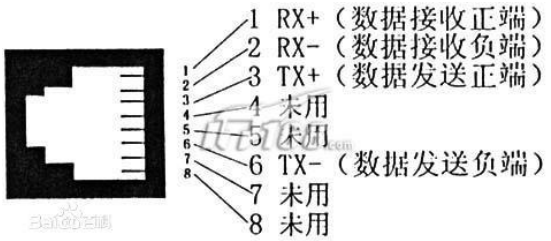


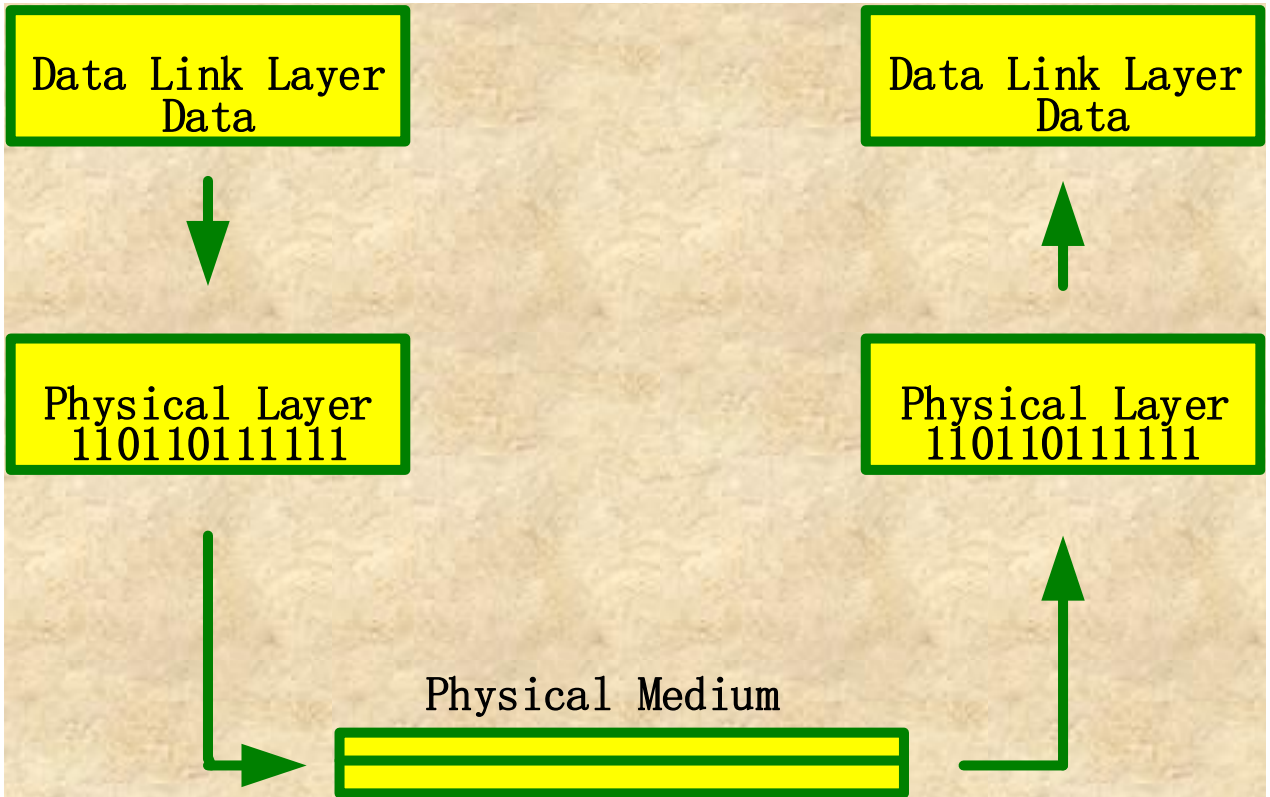
图 2：RJ45 型网线插头的 T568B 线序



# 2 Contents of Physical Layer Protocol

- Responsible for reliably transmitting bit data from one end of the physical medium to the other负责将比特数据从物理介质的一端可靠地传输到另一端



# 3 Basic theory of data communication

(1) Basic concepts

(2) For a specific physical communication channel, what is the maximum transmission rate? Infinite?

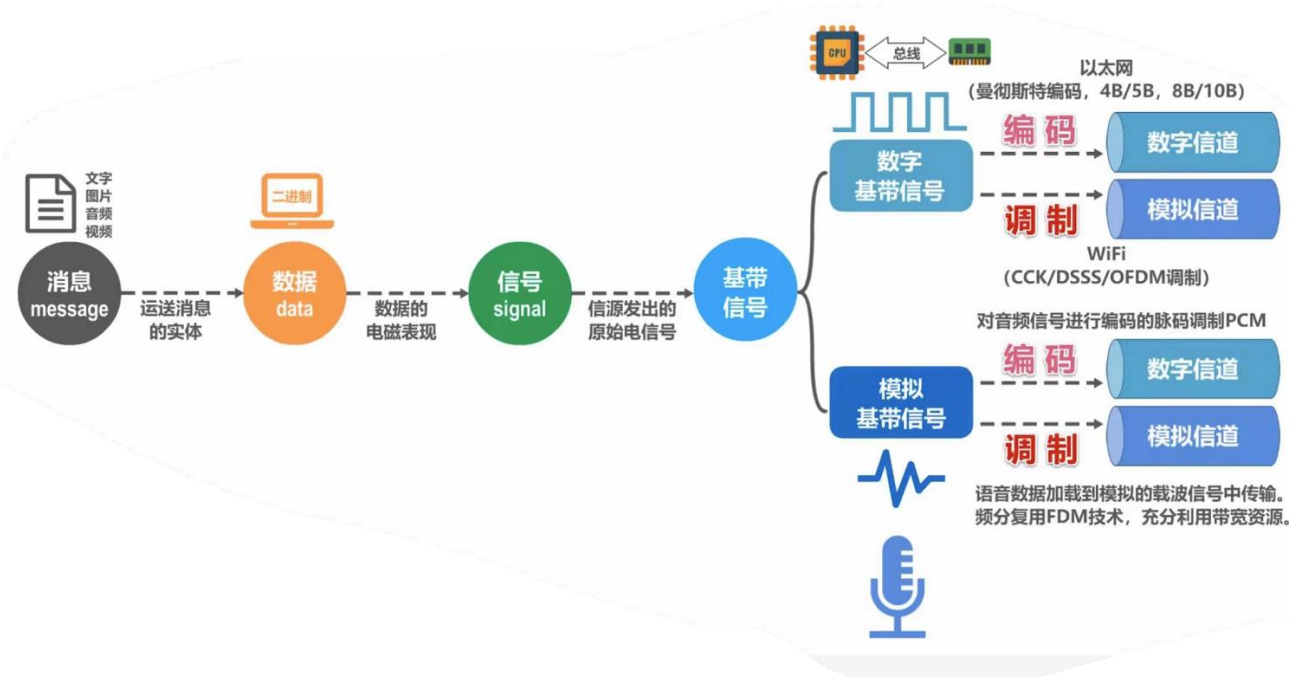
对于特定的物理通信信道，最大传输速率是多少？无限大？

(3) How to transmit Bits in specific physical communication channels? Transmission speed, transmission duration, signal distortion?如何在特定物理通信信道中传输比特？传输速度、传输持续时间、信号失真？

(4) In order to save communication equipment and costs, how to transmit the information of multiple computers in one physical channel?为了节省通信设备和成本，如何在一个物理信道中传输多台计算机的信息？

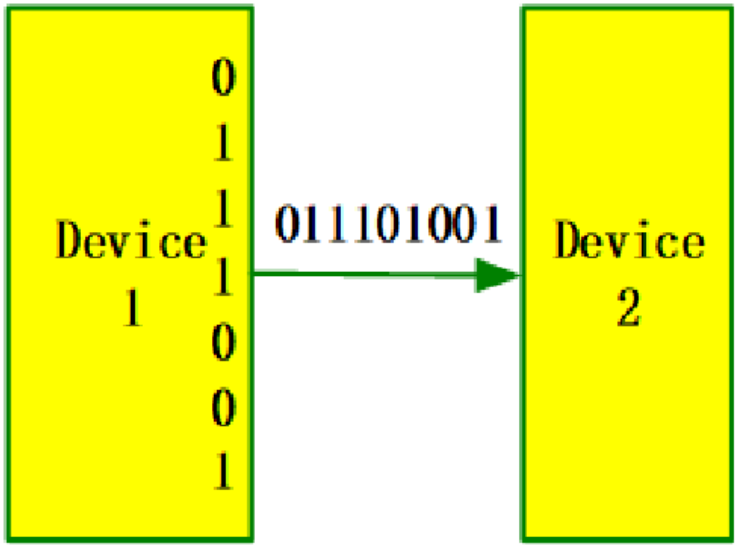
# (1) Basic concepts

- Data: Physical symbols describing objective facts
- 数据：描述客观事实的物理信号
- Signal: Form of data transmission process
- 信号：数据传输过程的形式
- Channel: media that sends information in a certain direction渠道：向特定方向发送信息的媒介



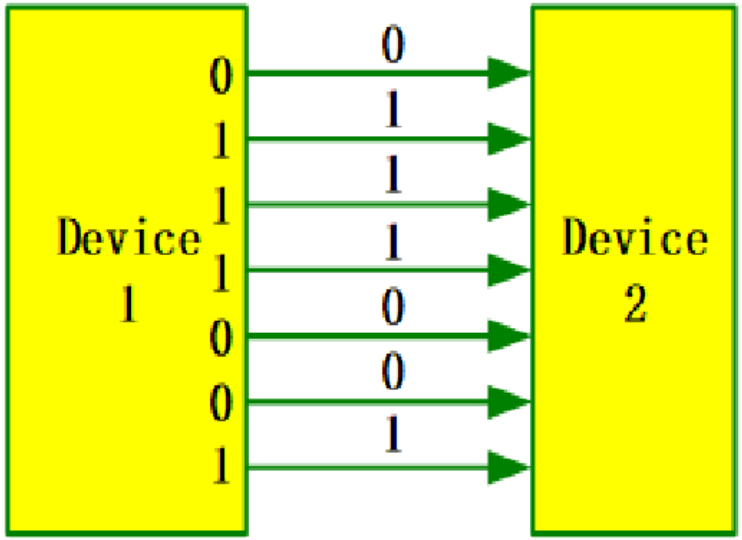
# (1) Basic concepts

## Serial transmission串行传输



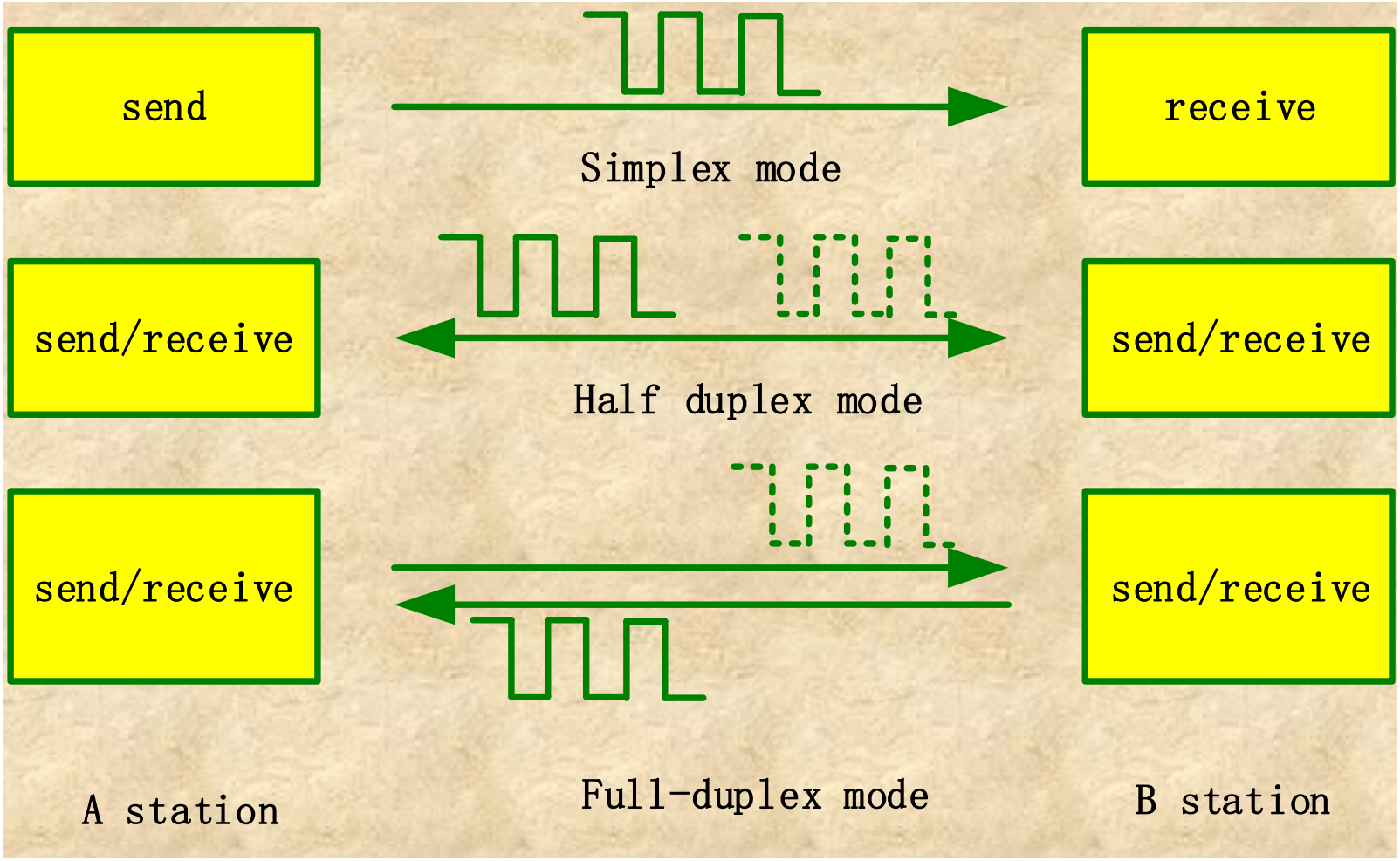

- Suitable for long distance communication适合长距离

## Parallel transmission并行传输




- Suitable for short distance communication适合短距离

# (1) Basic concepts



# 3 Basic theory of data communication

(1) Basic concepts

(2) For a specific physical communication channel, what is the maximum transmission rate? Infinite?

(3) How to transmit Bits in specific physical communication channels? Transmission speed, transmission duration, signal distortion?

(4) In order to save communication equipment and costs, how to transmit the information of multiple computers in one physical channel?

# (2) Channel Characteristics

- Symbol: A digital pulse, which represents the basic waveform of a discrete value码元 数字脉冲，代表离散值的基本波形
  - ASCII code of letter 'A': 1000001, can be represented by 7 pulses/symbols例如A的ASCII码，脉冲二进制
- The amount of information carried by a symbol is determined by the number of discrete values taken by the symbol一个符号所承载的信息量取决于该符号所取离散值的数量
  - Two discrete values ("0" or "1"): 1 bit码元取0，1两个离散值，一个码元携带1bit信息
  - Four discrete values for a symbol: 2 bits
  - If N discrete values for a symbol of n (bits), then

$$n = \log_2 N$$

# (2) Channel Characteristics

- Baud rate (modulation rate or symbol rate) : 波特率
  - The number of times the signal changes every second
  - 信号每秒变化的次数，也称为调制速率、码元速率
    - 1 baud sends one symbol per second1波特为每秒传送1个码元
- Bit rate : 比特率
  - The number of bits transmitted per second每秒传送的二进制位数
- Relationship between them:两者的关系
  - If each symbol has n bits: bit rate (b/s) = baud rate * n
  - If each symbol has 3 bits, then the bit rate is three times the baud rate
  - If each symbol has 1 bit, the bit rate and baud rate are the same

# (2) Channel Characteristics

- Channel capacity (bandwidth in networks)信道容量
  - The maximum number of bits that can be transmitted over a channel per unit time, expressed in bps单位时间内信道上所能传输的最大bit数
- Frequency bandwidth (in communication)频带宽
  - Frequency range of the signal allowed by the channel (= maximum frequency - minimum frequency), in Hz 信道所允许它通过信号的频率范围
  - E.g., people can hear sound waves of 20Hz ~ 20kHz, i.e., frequency bandwidth of hearing system is19980Hz

# (2) Channel Characteristics

- Nyquist's Law :奈奎斯特

  1) For an ideal low-pass channel with the frequency bandwidth of W (Hz), its maximum symbol transmission rate = 2W Baud理想低通信道下最高码元传输速率=2w Baud

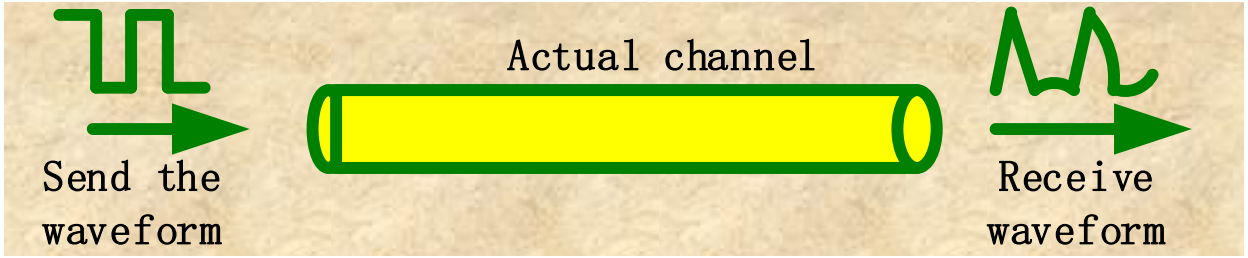
  2) For an ideal band-pass channel with the frequency bandwidth of W (Hz), its maximum symbol transmission rate = W Baud理想带通信道

- Question
  - The frequency band of a standard telephone channel is 300 ~ 3400Hz, i.e., the frequency bandwidth is 3100Hz. What is the maximum symbol transmission rate through the ideal low-pass channel?
  - 任何实际信道都不是理想的，在传输信号时会产生各种失真以及带来多种干扰。码元传输的速率越高，或信号传输的距离越远，在信道的输出端的波形的是真就越严重

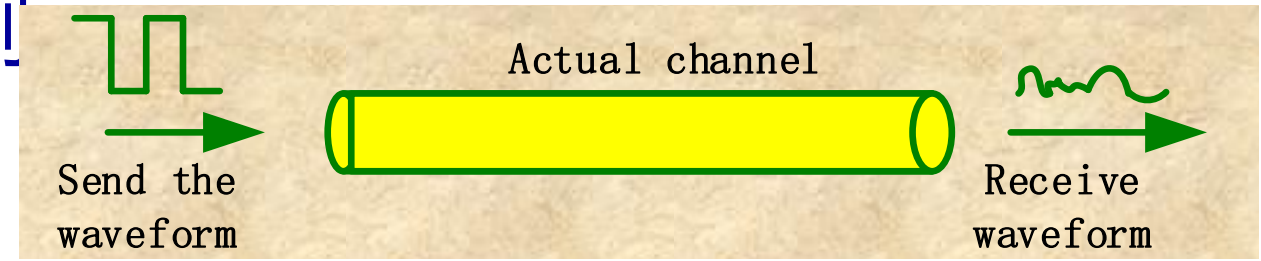
# (2) Channel Characteristics

- Actual channels are not ideal – Distortion exists
  - Higher symbol transmission rate高速
  - Farther signal transmission distance远距

  → Larger distortion
- Small and recognizable distortion有失真，但可识别




- Large and unrecognizable distortion失真大，无法识别



# (2) Channel Characteristics

- Nyquist's Law :

  1) For an ideal low-pass channel with the frequency bandwidth of W (Hz), its maximum symbol transmission rate = 2W Baud

  2) For an ideal band-pass channel with the frequency bandwidth of W (Hz), its maximum symbol transmission rate = W Baud

- Question
  - The frequency band of a standard telephone channel is 300 ~ 3400Hz, i.e., the frequency bandwidth is 3100Hz. What is the maximum symbol transmission rate through the ideal low-pass channel?

# (2) Channel Characteristics

- Shannon's Formula: limit information transmission rate C of channel can be expressed as:信道的极限信息传输速率C可表示为

$$C = W\log_2(1+S/N)$$

  - W - Frequency bandwidth in Hz信道频带宽
  - S - Average signal power through the channel信道内所传信号的平均功率
  - N - Gaussian noise power through the channel信道内部高斯噪声功率
  - S/N – related to signal-to-noise ratio信噪比 $10\log_{10}S/N$
    - Typically in db (decibels):
- Actual rate much lower than C due to signal loss
- Possible to achieve error-free transmission as

# (2) Channel Characteristics

- Example:
  - For the standard telephone channel with frequency bandwidth of 3.1 kHz, if the signal-to-noise ratio S/N = 2500, then can the information transmission rate be 50kb/s?
- Solution:
  - Put W = 3.1kHz, S / N = 2500 into Shannon's Formula
  - The limit information transmission rate is 35kb/s, so it is impossible to reach 50kb/s
- In order to achieve 50kb/s, you can
  - improve the signal-to-noise ratio in the channel
  - or increase the frequency bandwidth

# 3 Basic theory of data communication

(1) Basic concepts

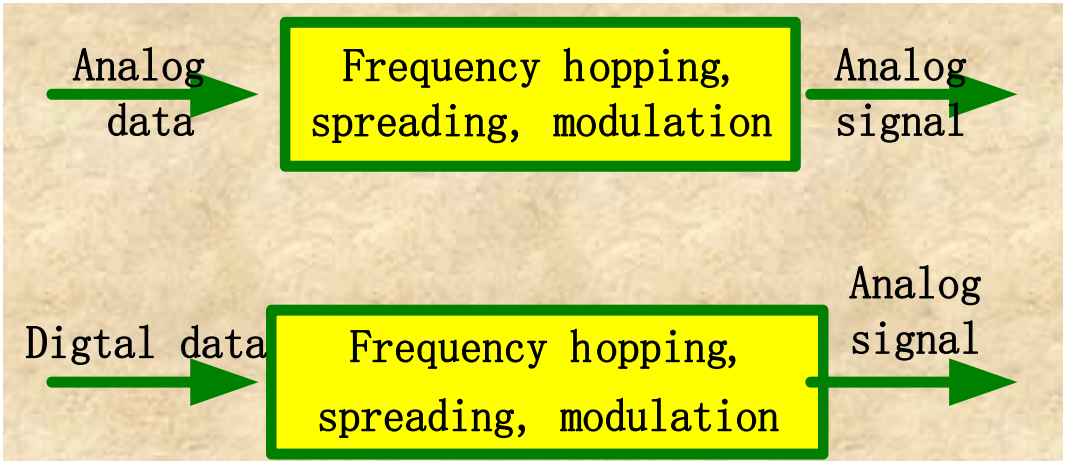
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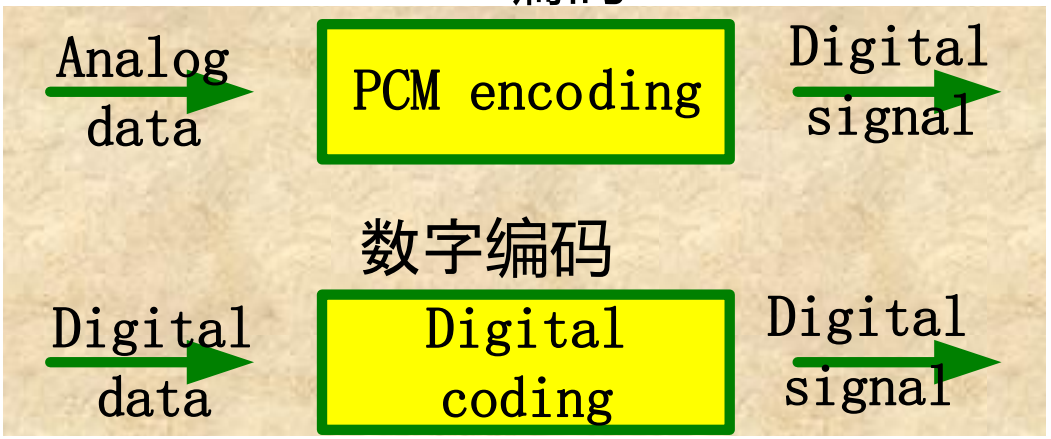
(4) In order to save communication equipment and costs, how to transmit the information of multiple computers in one physical channel?

# Techniques in analog/digital transmission

跳频、扩频、调制



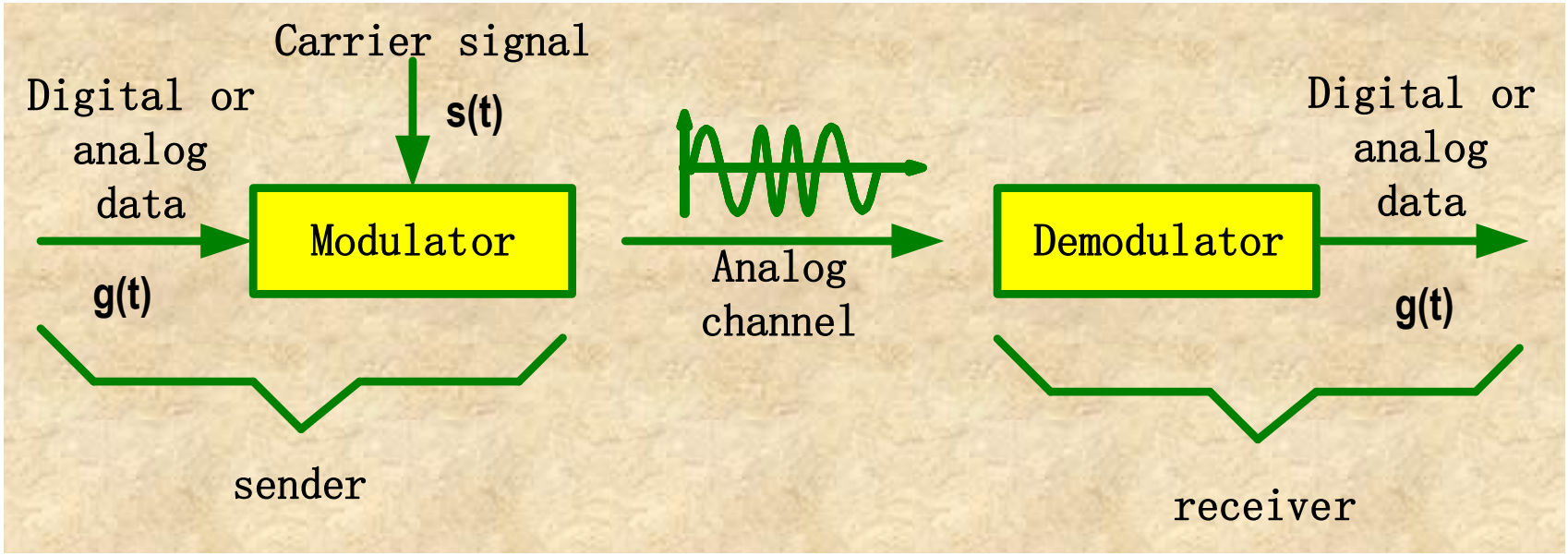

PCM编码



# Modulation / demodulation system model

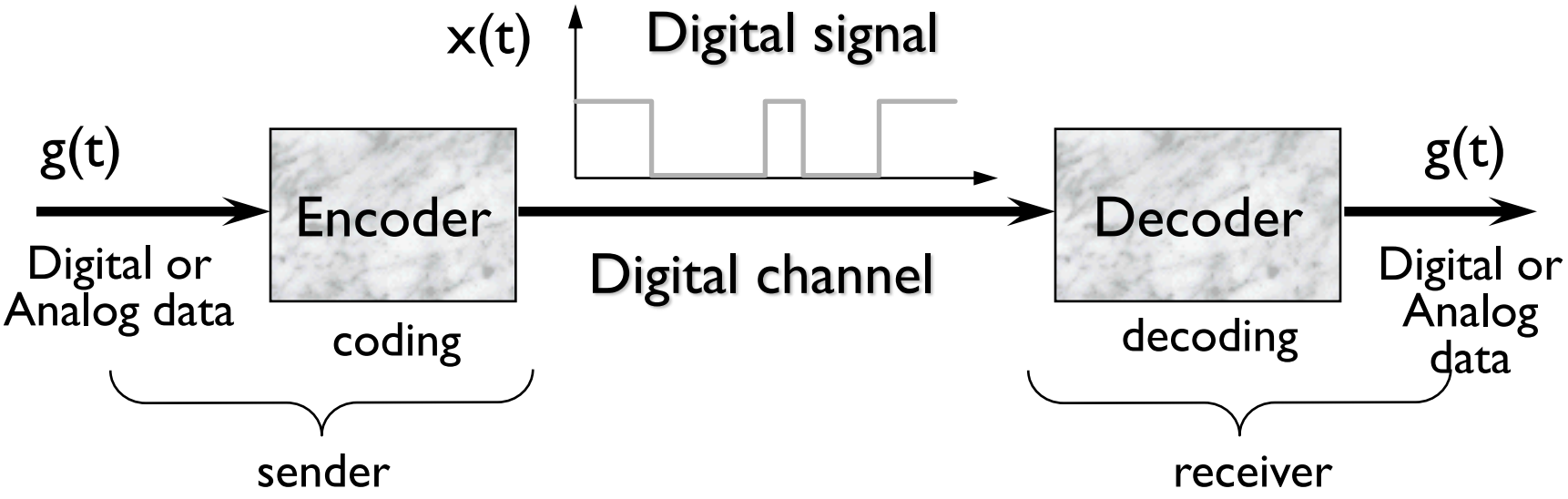
调制/解调系统模型

载波信号



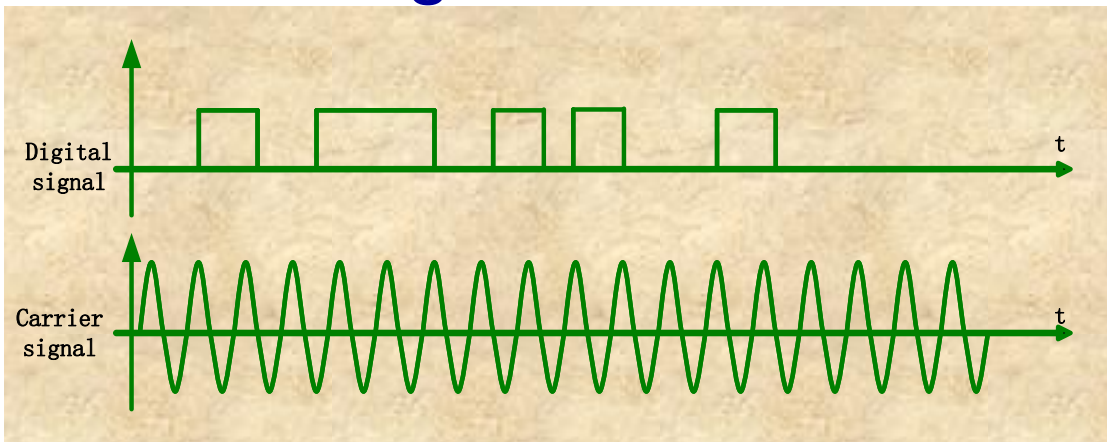
# Coding / decoding system model

编码/解码系统模型

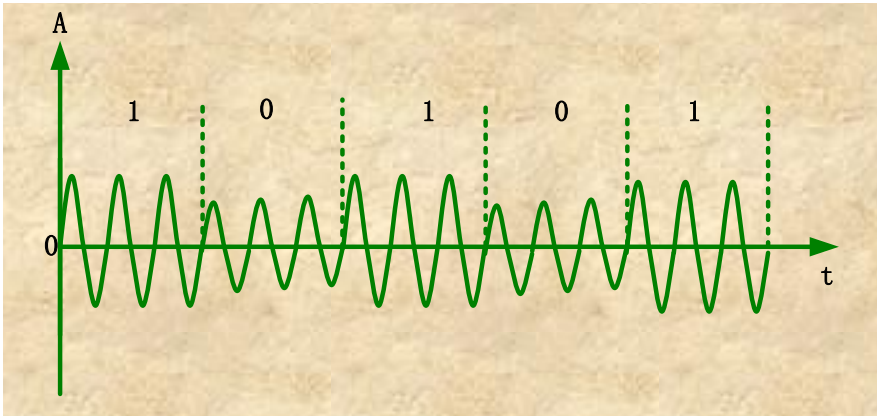


# Modulation 调制技术 (digital data to analog signals)

- Digital and carrier signals

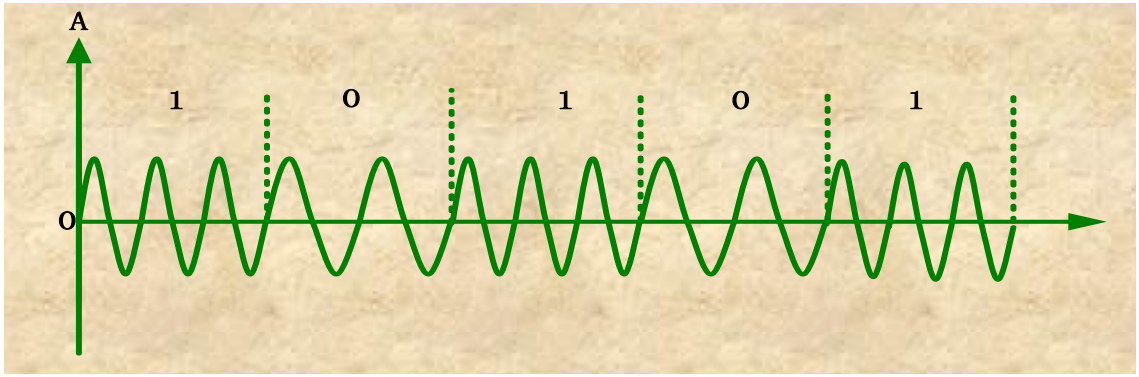



- Amplitude Modulation (AM)调幅，载波的振幅随基带数字信号变化
- （ASK）

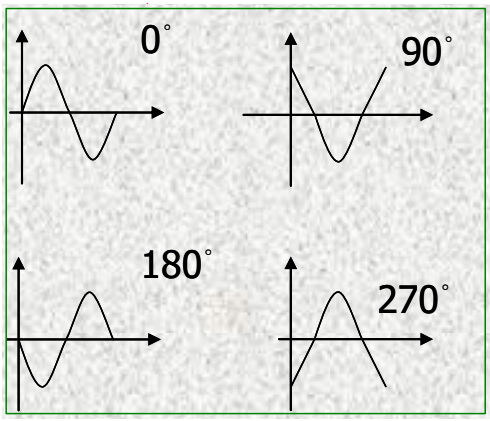


# Modulation
## (digital data to analog signals)

- Frequency Modulation (FM)调频，载波频率随基带数字信号变化（FSK）




- Phase Modulation (PM)调相，载波初始相位…（

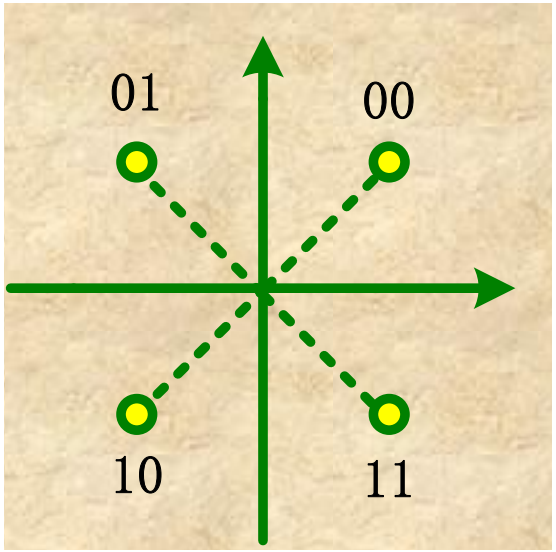




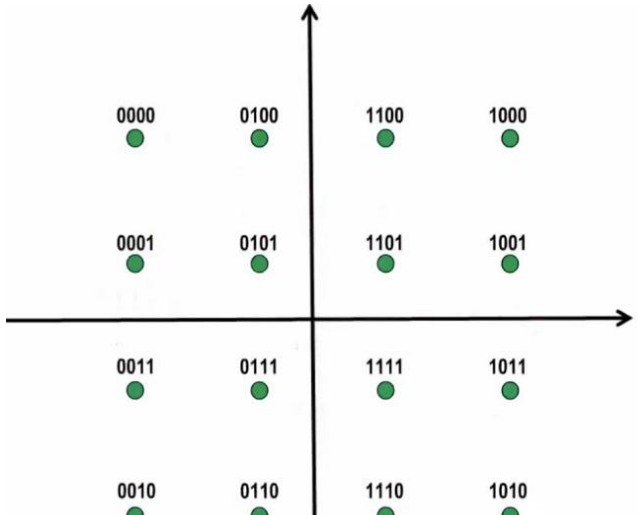

# Modulation
## (digital data to analog signals)

- Quadrature Amplitude Modulation (QAM)正交条幅QAM技术
  - PM and AM techniques结合调相和调幅技术
  - If there are x changes in phase and y changes in amplitude, then there are x * y combined changes
  - QAM星座表

4-QAM constellation table

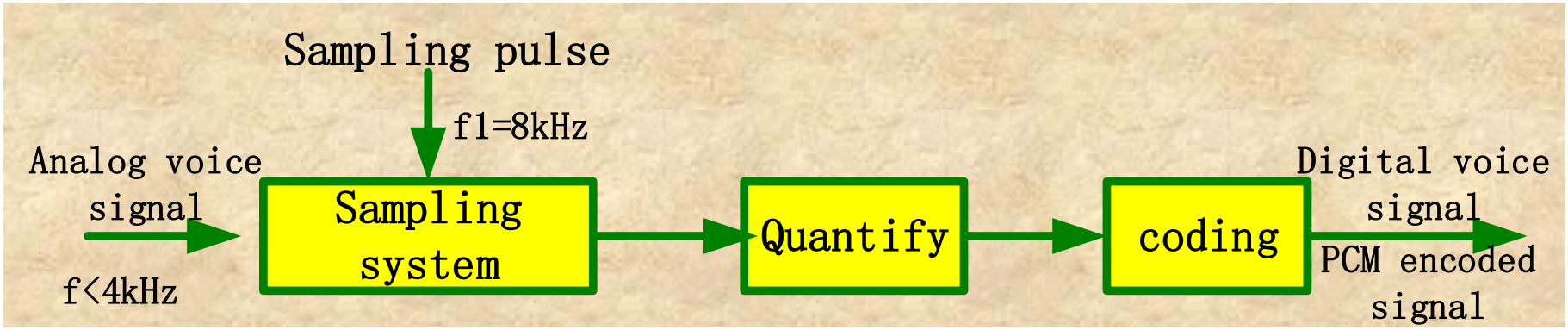



16-QAM constellation table



# Pulse Code Modulation (PCM)脉冲编码调制，PCM技术 (analog data to digital signals)

- Sampling Theorem
  - As long as the sampling frequency is not less than twice the maximum frequency of the signal, the original signal can be recovered from the sampling pulse without distortion
- PCM system model采样、量化、编码

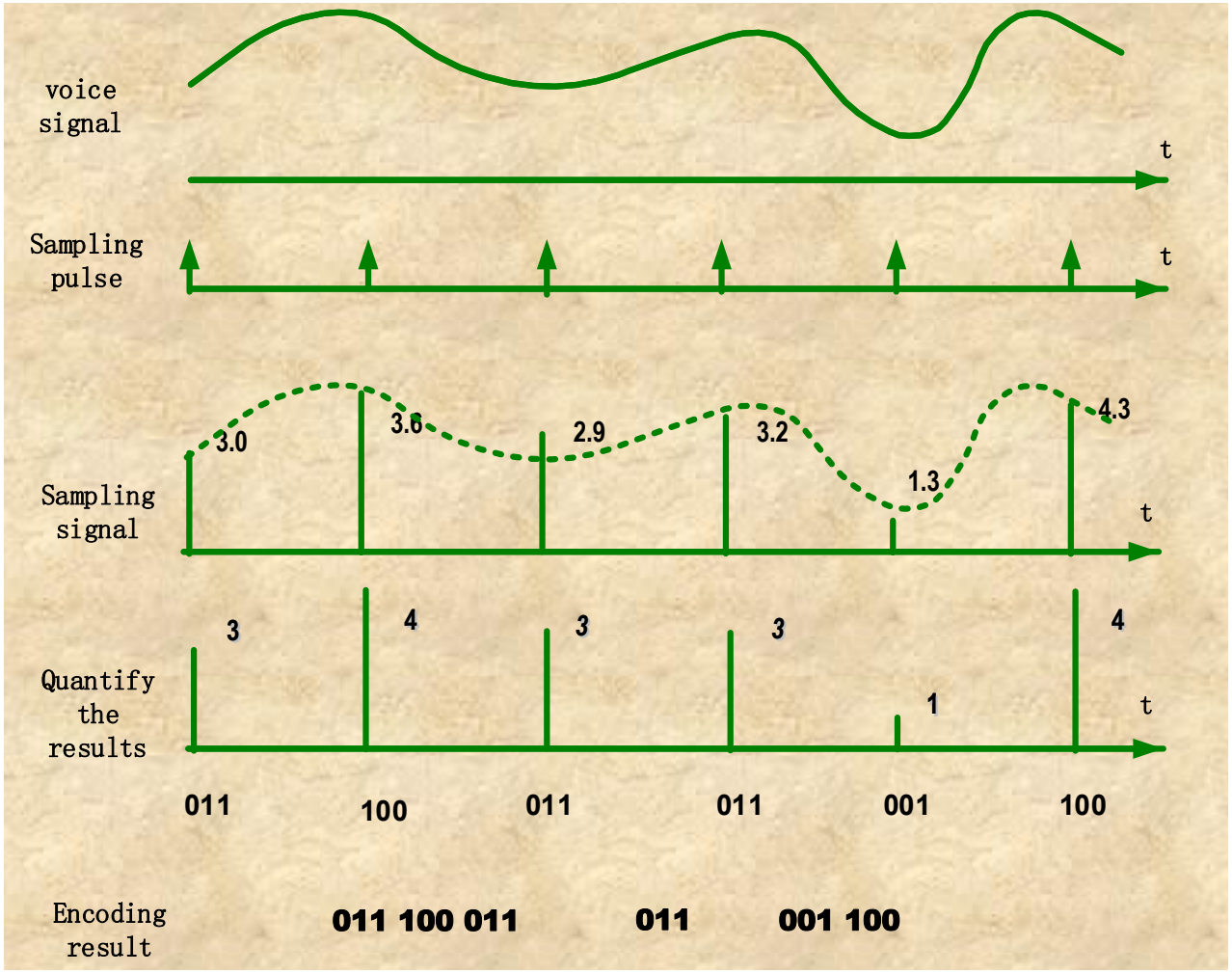


# Pulse Code Modulation (PCM) (analog data to digital signals)

- Voice signal digitization语音信号数字化
  - Voice frequency bandwidth f <4kHz
  - Sampling clock frequency: 8kHz (> 2 times the maximum voice frequency)
  - Sample quantization series: 256 (8bit / per sample)
  - Data rate: 8000 times / s * 8bit = 64kb / s
  - The rate of each PCM signal = 64kb / s

# Pulse Code Modulation (PCM)
## (analog data to digital signals)

PCM coding process example

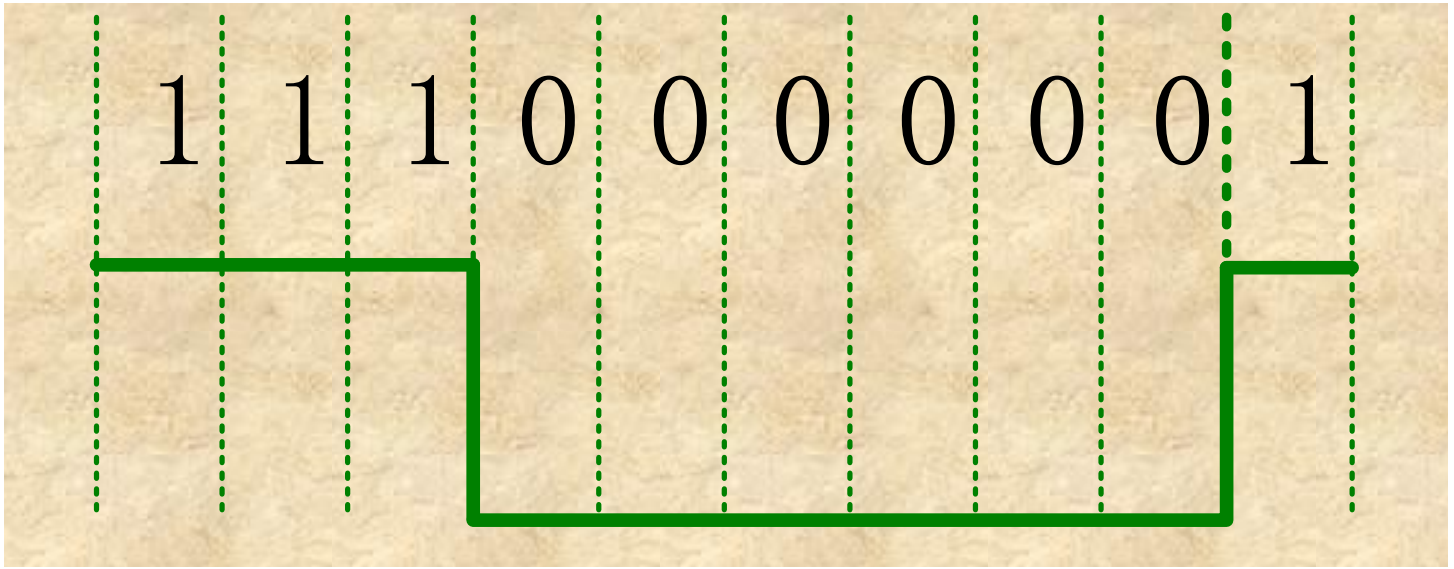


# Digital signal coding
## (digital data to digital signals)

- Non-Return-to-Zero (NRZ)不归零编码
- Manchester encoding曼彻斯特编码
- Differential Manchester encoding差分曼彻斯特编码
- Block encoding (4B / 5B, 8B / 10B)块编码

# Digital signal coding
## (digital data to digital signals)

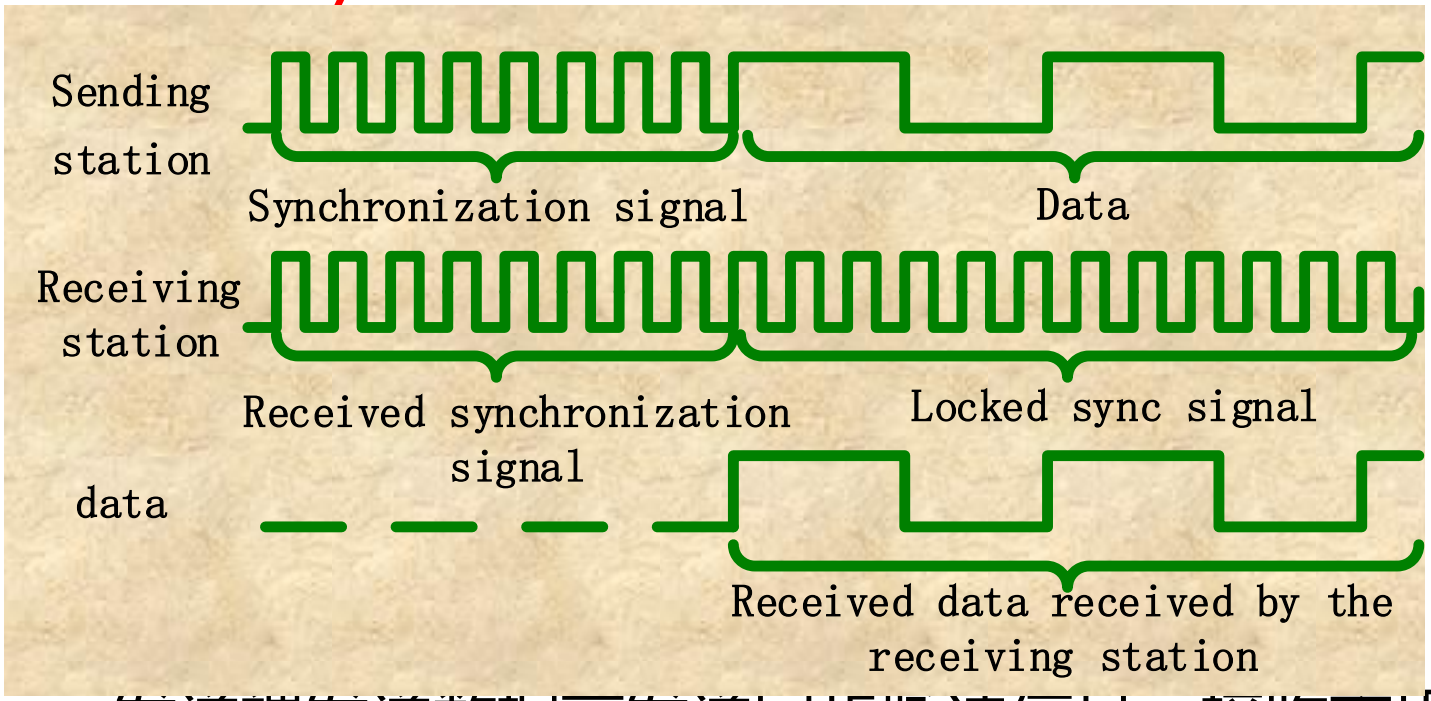
- Non-Return-To-Zero Level (NRZ-L) Coding
- 低电平表示二进制0，高电平表示二进制1



# Digital signal coding (digital data to digital signals)

- Disadvantages of NRZ-L
  - Difficult to tell the end of one and the beginning of another难以分辨一位的结束和另一位的开始
  - The sender and the receiver must have clock synchronization发送方和接收方必须有时钟同步
  - If the signal "0" or "1" appears continuously, the signal DC component will accumulate0，1连续出现时，直流分量会累加
  - Prone to propagation errors容易产生传播错误

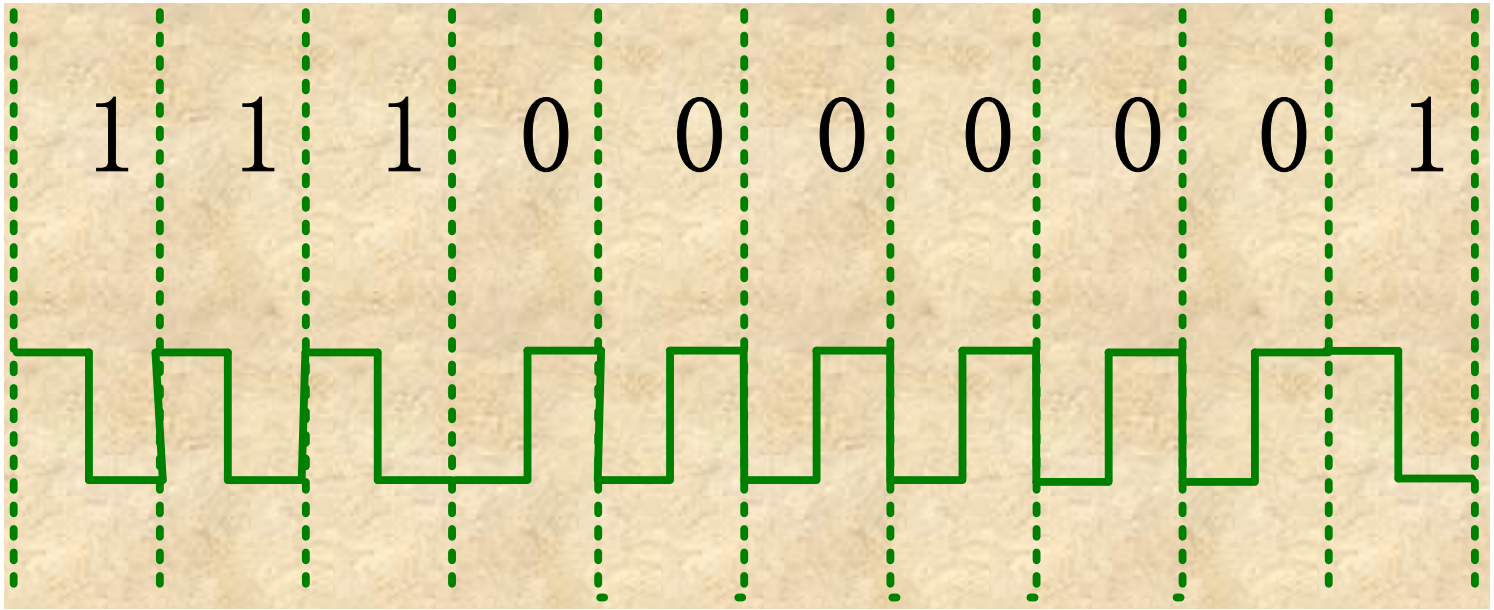
- Bit synchronization位同步：
  - The purpose is to synchronize each bit of information received by the receiver with the sender接收端接受的每一位消息都与发送端同步
- External synchronization




  - 发送端发送数据前发送同步脉冲信号，接收方用接收到的同步信号锁定时钟频率
- Self-synchronization
- 通过特殊编码使数据编码信号中包含同步信号，接收

# Digital signal coding (digital data to digital signals)

- Manchester Coding
  - Each symbol is divided into two equal intervals
    - "1": the previous interval is high and the next low
    - "0": the previous interval is low and the next high
  - Feature: A level hop in the middle time of each symbol



# Digital signal coding 数字编码技术 (digital data to digital signals)

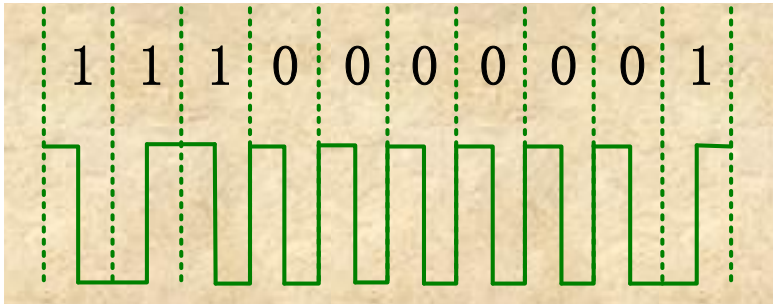
- Advantages
  - Overcoming the disadvantages of the NRZ Coding
  - Each jump in the middle of a symbol can be used as data, and also as a clock for self-synchronization
- Disadvantages
  - Doubling the frequency bandwidth requirement of the transmission channel
  - High-frequency noise also increases – susceptible to noise interference
  - Ambiguous

# Digital signal coding (digital data to digital signals)

- Differential Manchester Coding
  - "1": the level of the first half of the symbol is the same as the level of the second half of the previous symbol
  - "0": the level of the first half of the symbol is opposite to the level of the second half of the previous symbol
- Regardless of whether the previous bit is 1 or 0

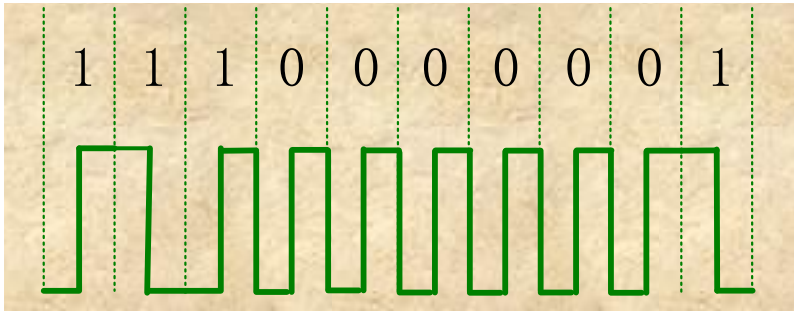
Waveform 1

Assuming the previous bit rises from low to high




Waveform 2

Assuming the previous bit drops from high to low



# Digital signal coding (digital data to digital signals)

- Advantages
  - Regardless of whether the symbol is "1" or "0", there must be a level hop in the middle of each symbol => better anti–interference performance
  - Separation between bit clock and data => easy data extraction
  - Transition uniquely determined by the second half of the previous symbol => ambiguity eliminated
- Disadvantages
  - Doubling the frequency bandwidth requirement of the transmission channel
  - High-frequency noise also increases – susceptible to noise interference

# Digital signal coding
# (digital data to digital signals)

- Exercise:

  Draw the waveform of "0011101" with
  - NRZ-L
  - Manchester Coding
  - Differential Manchester Coding

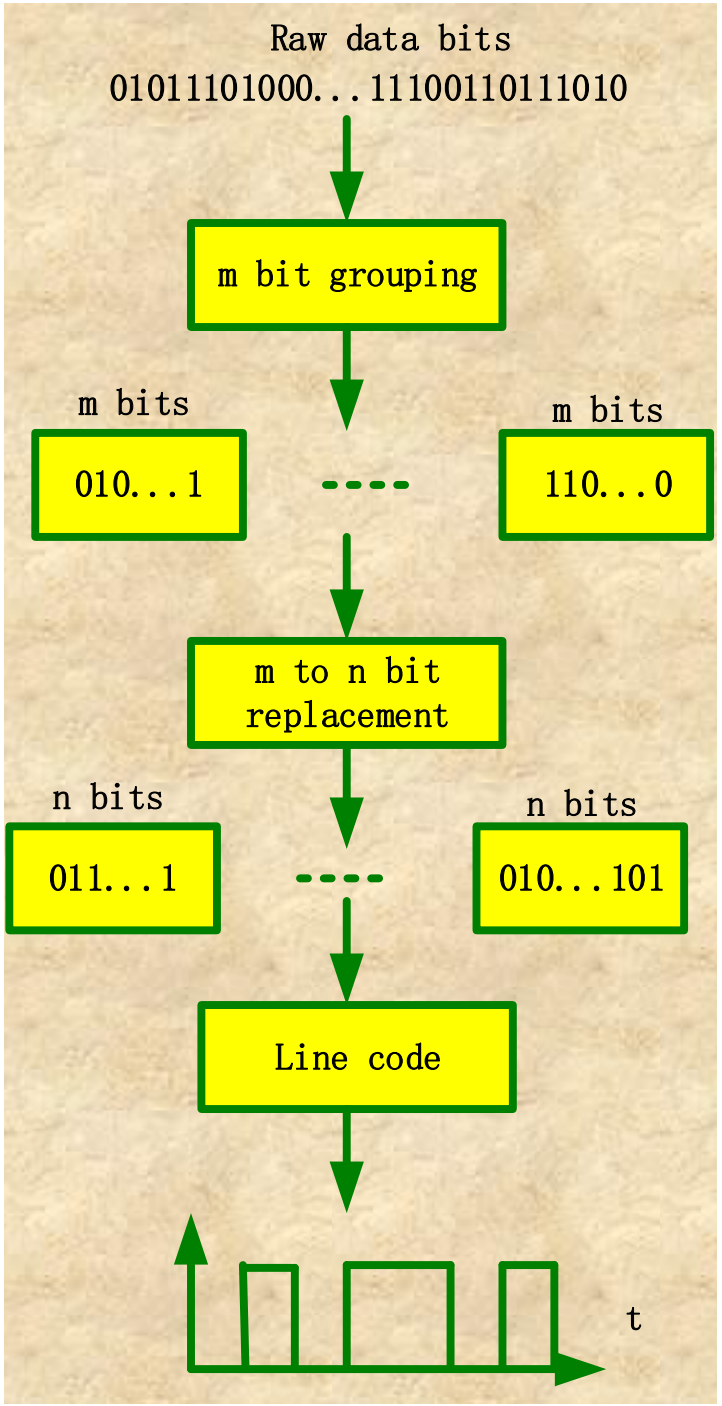
# Digital signal coding (digital data to digital signals)

- Block Coding
  - Advantages
    - Can improve the coding efficiency, reduce the modulation rate, and reduce the transmission line requirements可以提高编码效率，降低调制速率，降低对传输线路的要求
    - Some redundant bits can be added for error detection or synchronization可增加一些冗余位，用于检错或同步。
  - Applications
    - 4B / 5B (FDDI, 100M Fast Ethernet)
    - 8B / 10B (Gigabit Ethernet)

# Digital signal coding (digital data to digital si

Steps (mB / nB, m < n):

- Grouping 分组，数据比特序列划分为m位的多个比特序列
- Substitution置换
- n位置换m位，多出的比特序列可以用来检错、同步或其他控制功能
  - The extra bit sequence can be used for
    - error detection
    - synchronization
    - or other controls



# 3 Basic theory of data communication

(1) Basic concepts

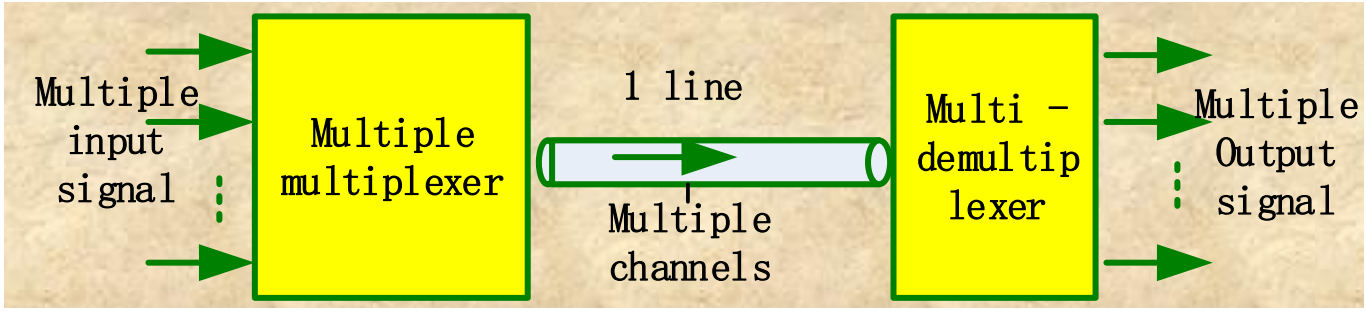
(2) For a specific physical communication channel, what is the maximum transmission rate? Infinite?

(3) How to transmit Bits in specific physical communication channels? Transmission speed, transmission duration, signal distortion?

(4) In order to save communication equipment and costs, how to transmit the information of multiple computers in one physical channel?

# Multiplexing (Multi-pass)多路复用技术/多路共传

- To send multiple sources of information in one physical channel at the same time一条物理通道传多路信息

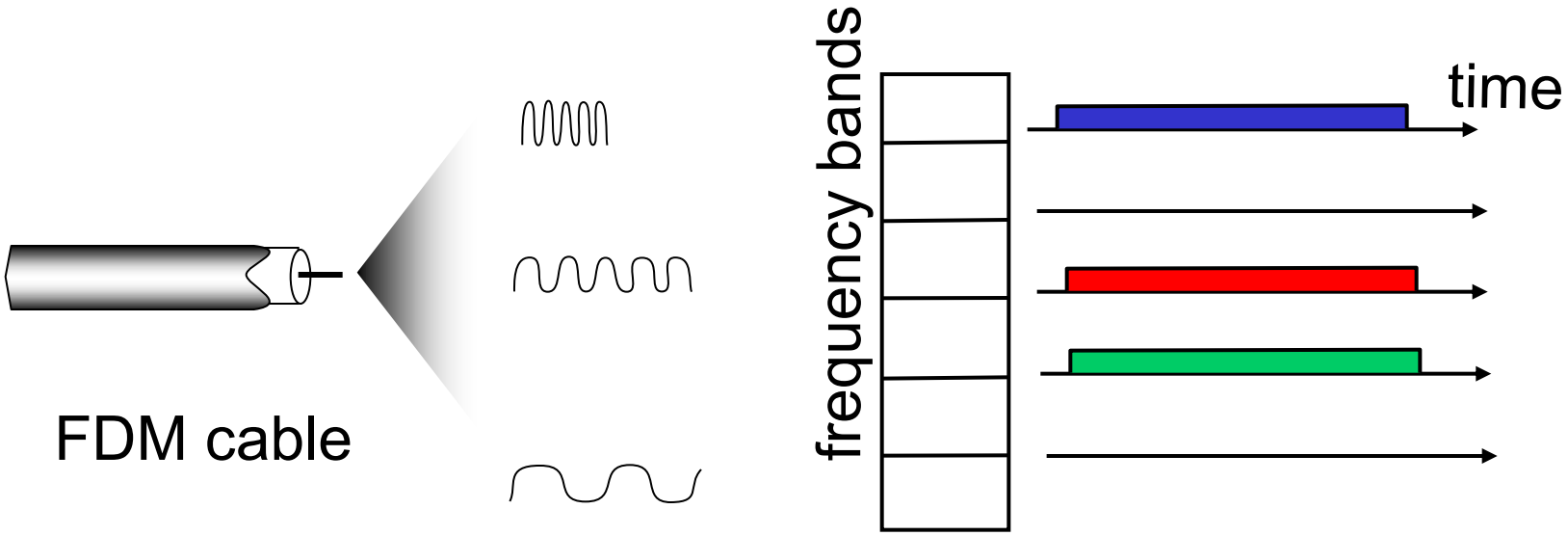



- Commonly used techniques
  - Frequency division multiplexing (FDM) 频分复用
  - Time division multiplexing (TDM)时分复用
  - Wavelength division multiplexing (WDM)波分复用
  - Code division multiplexing (CDM)码分复用

# *Multiplexing : FDMA

## FDMA: frequency division multiple access

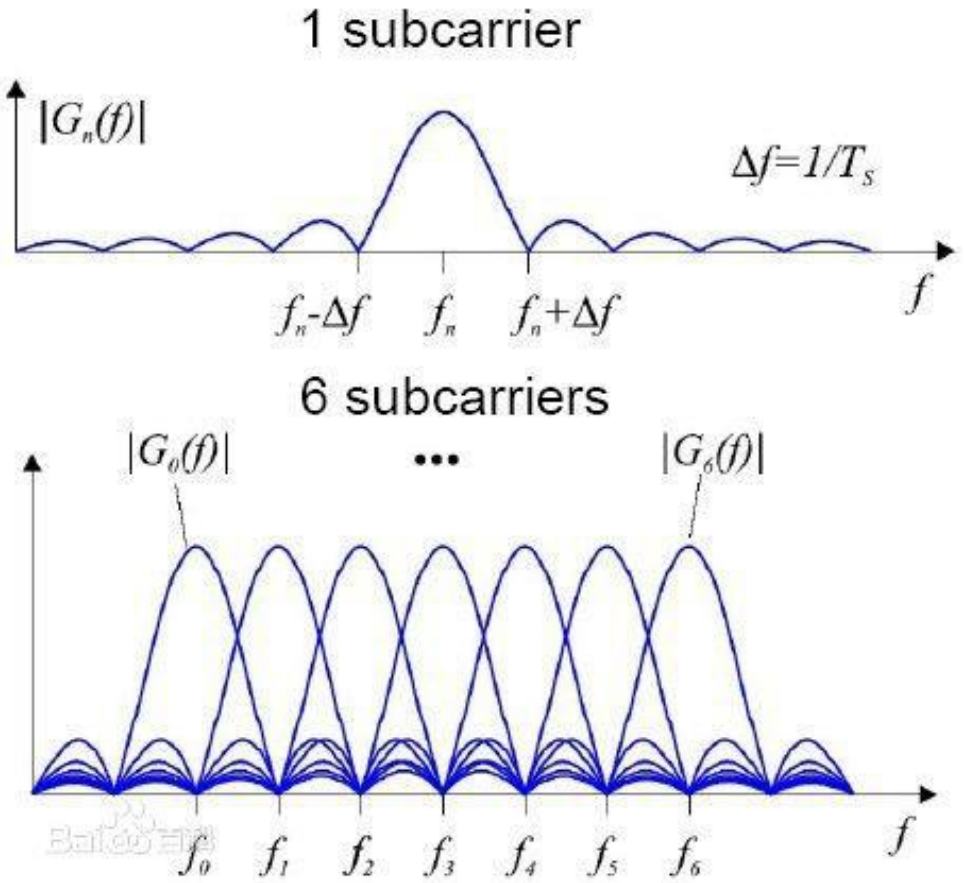
- channel spectrum divided into frequency bands
- each station assigned fixed frequency band
- unused transmission time in frequency bands go idle
- example: 6-station LAN, 1,3,4 have pkt, frequency bands 2,5,6 idle



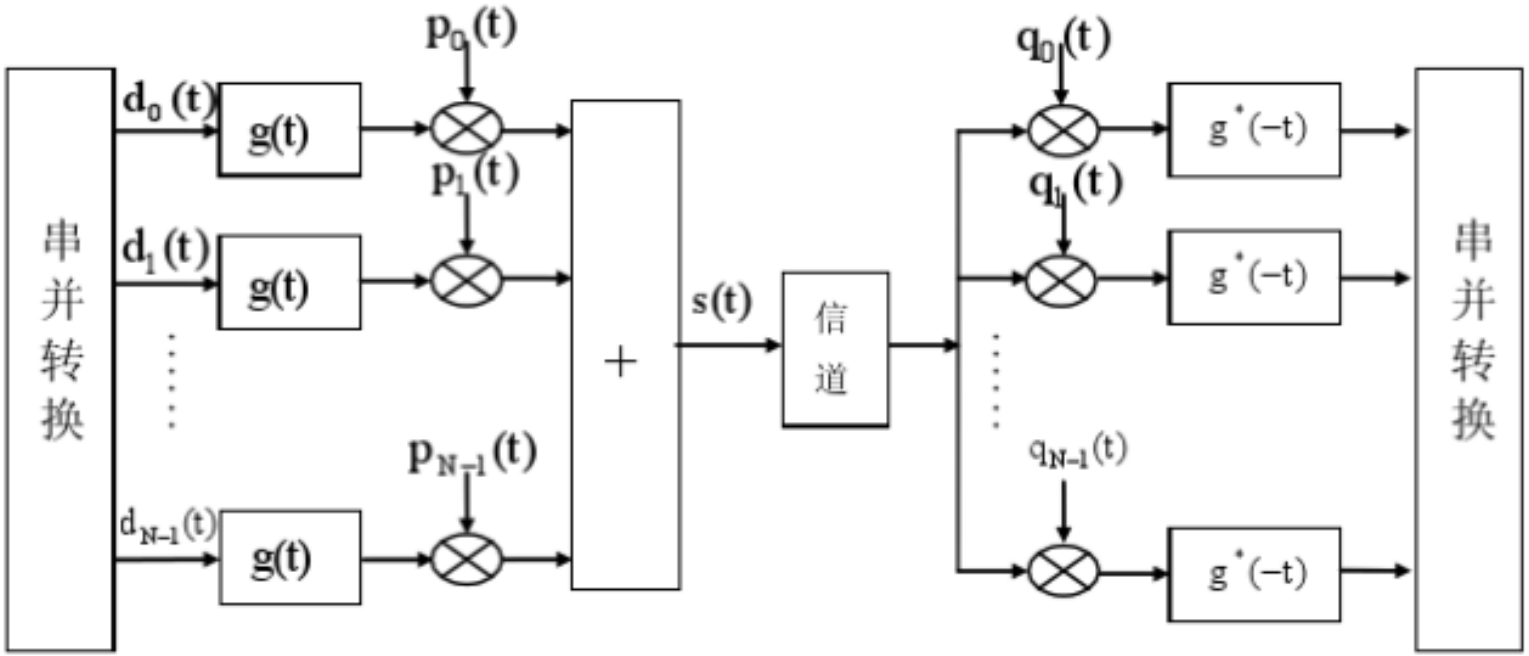
# *Multiplexing : Orthogonal frequency division multiplexing (OFDM) technology

- Basic principle:
  - Dividing the channel into a number of orthogonal sub-channels;将多路不同信号的频率调制到不同的频率范围。
  - The high-speed serial data is decomposed into multiple parallel low-speed data, the use of multi-carrier FDM method to transfer, that is, the data allocated to a large number of sub-channels for transmission.将高速串行数据分解成多个并行的低速数据，利用多载波 FDM 方法进行传输，即把数据分配给大量子信道进行传输。
- Suitable for the high-speed data transmission through wireless channel under the presence of multipath propagation and Doppler shift.适合在存在多径传播和多普勒频移的情况下，通过无线

# *Multiplexing : Orthogonal frequency division multiplexing (OFDM) technology



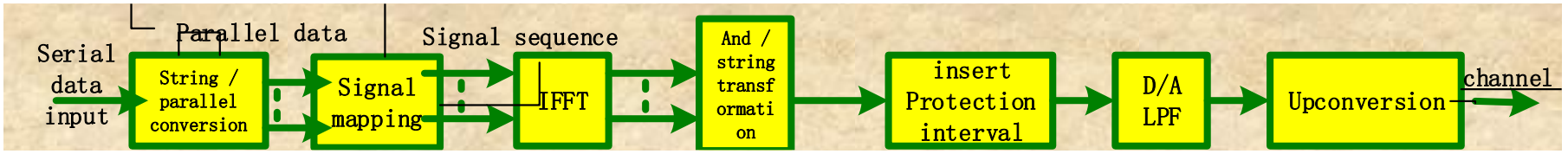
# *Multiplexing : Orthogonal frequency division multiplexing (OFDM) technology



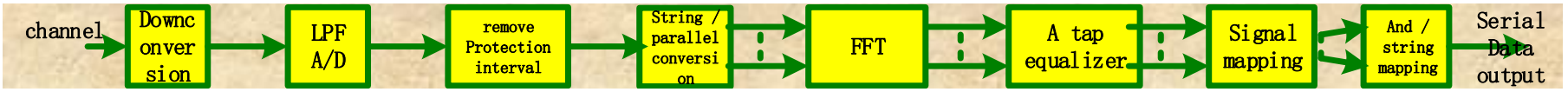

$$\int_0^{T_s} P_m(t)q_m(t)dt = c\delta_{mn} = \begin{cases} 0 & m \neq n \\ C & m = n \end{cases}$$

# *Multiplexing : Orthogonal frequency division multiplexing (OFDM) technology

- FFT based OFDM Transmit System




- FFT based OFDM Receiving System



# *Multiplexing : Orthogonal frequency division multiplexing (OFDM) technology

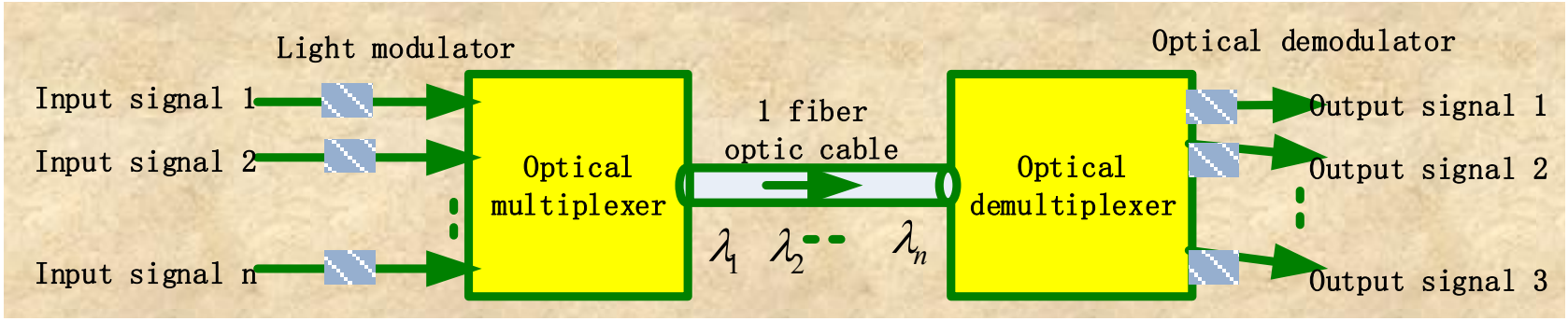
- Advantages:
  - Reduce the influence of inter-symbol interference;
  - Reduce the effect of frequency selective fading;
  - Improve the band utilization, to avoid the crosstalk between sub-channels;
  - Nearly eliminate inter-symbol interference;
  - Anti-interference coding technology can be used to effectively recover the error;
  - The parallel data can be modulated and demodulated by discrete Fourier transform DFT, which reduces the complexity of system implementation.

# *Multiplexing : Orthogonal frequency division multiplexing (OFDM) technology

- Applications of OFDM technology:
  - ADSL
  - Digital Audio Broadcasting (DAB)
  - HDTV
  - Wireless LAN (WLAN)
  - Broadband radio access network
  - 3G mobile communication network and other fields.

# *Multiplexing : Wavelength division multiplexing (WDM)

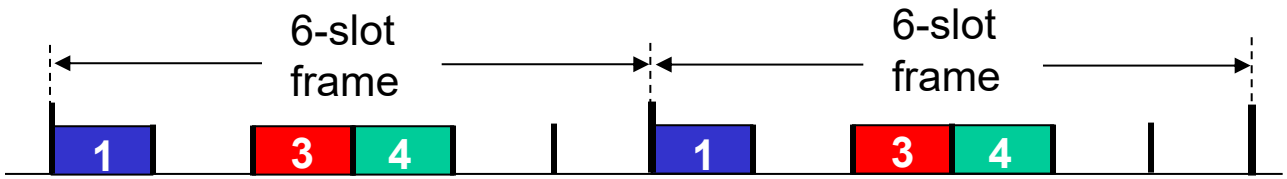
- Transmission of multiple optical carriers with different wavelengths in a fiber at the same time
- WDM is actually a variant of FDM for fiber channel multiplexing.

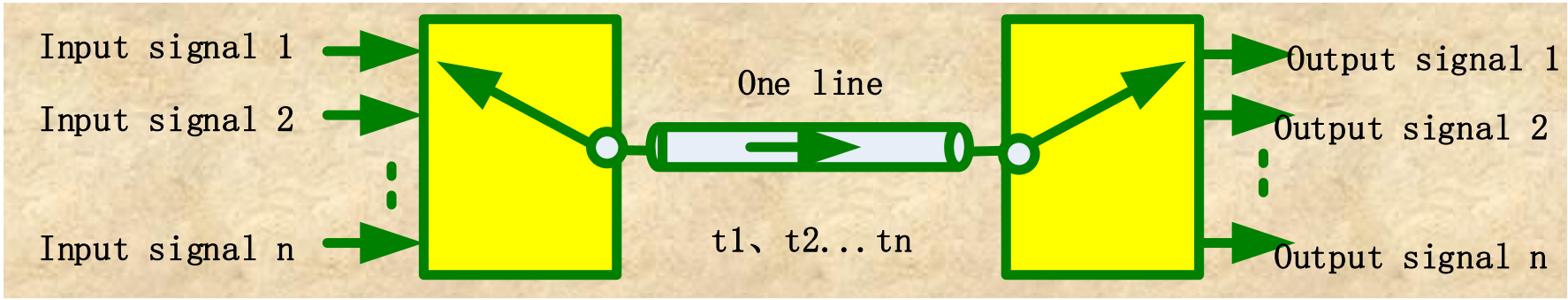


# *Multiplexing : TDMA

## TDMA: time division multiple access

- access to channel in "rounds"
- each station gets fixed length slot (length = pkt trans time) in each round
- unused slots go idle
- example: 6-station LAN, 1,3,4 have pkt, slots 2,5,6 idle



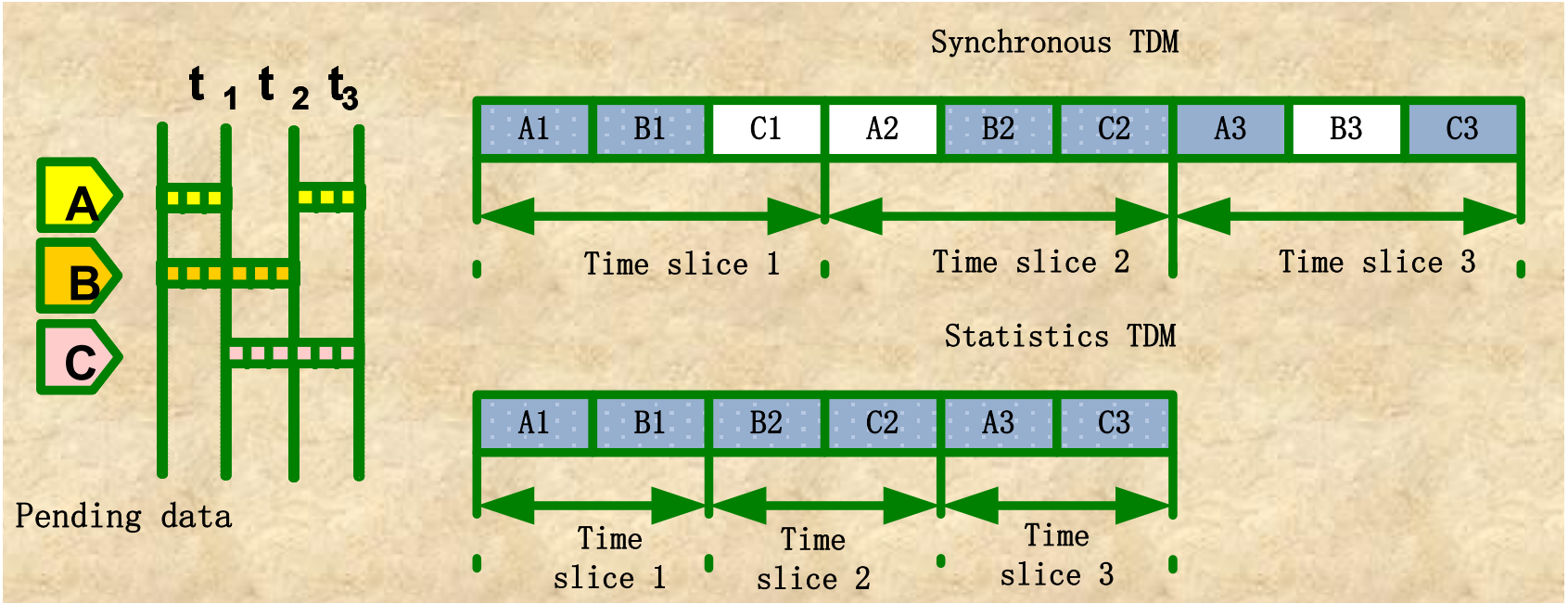
# *Multiplexing : TDMA

- Classification of TDM
  - Synchronous TDM:
    - Time slice fixed allocation, suitable for fixed rate transmission
    - The time slice of the physical channel is fixedly allocated to several users for data transmission, and each user uses the channel for data transmission when its corresponding time slice arrives.
  - Advantages: **simple implementation.**
  - Disadvantages: **there is a waste of bandwidth.**
  - **Synchronous TDM technology is suitable for fixed rate data communication system.**

# *Multiplexing : TDMA

- Asynchronous TDM:
  - The time slot of the physical channel is fixedly allocated for several, and the user does not occupy a certain time slice.
  - When a user needs to carry out data transmission, to assign a time slice;
  - If the user does not have the data transmission requirements, the system does not allocate the time slice to the user, the corresponding time slice can be allocated to other users.
- Advantages: **On-demand distribution of channel time slices, high utilization rate.**
- Disadvantages: **complex implementation.**
- **Asynchronous TDM technology is suitable for variable rate communication systems.**

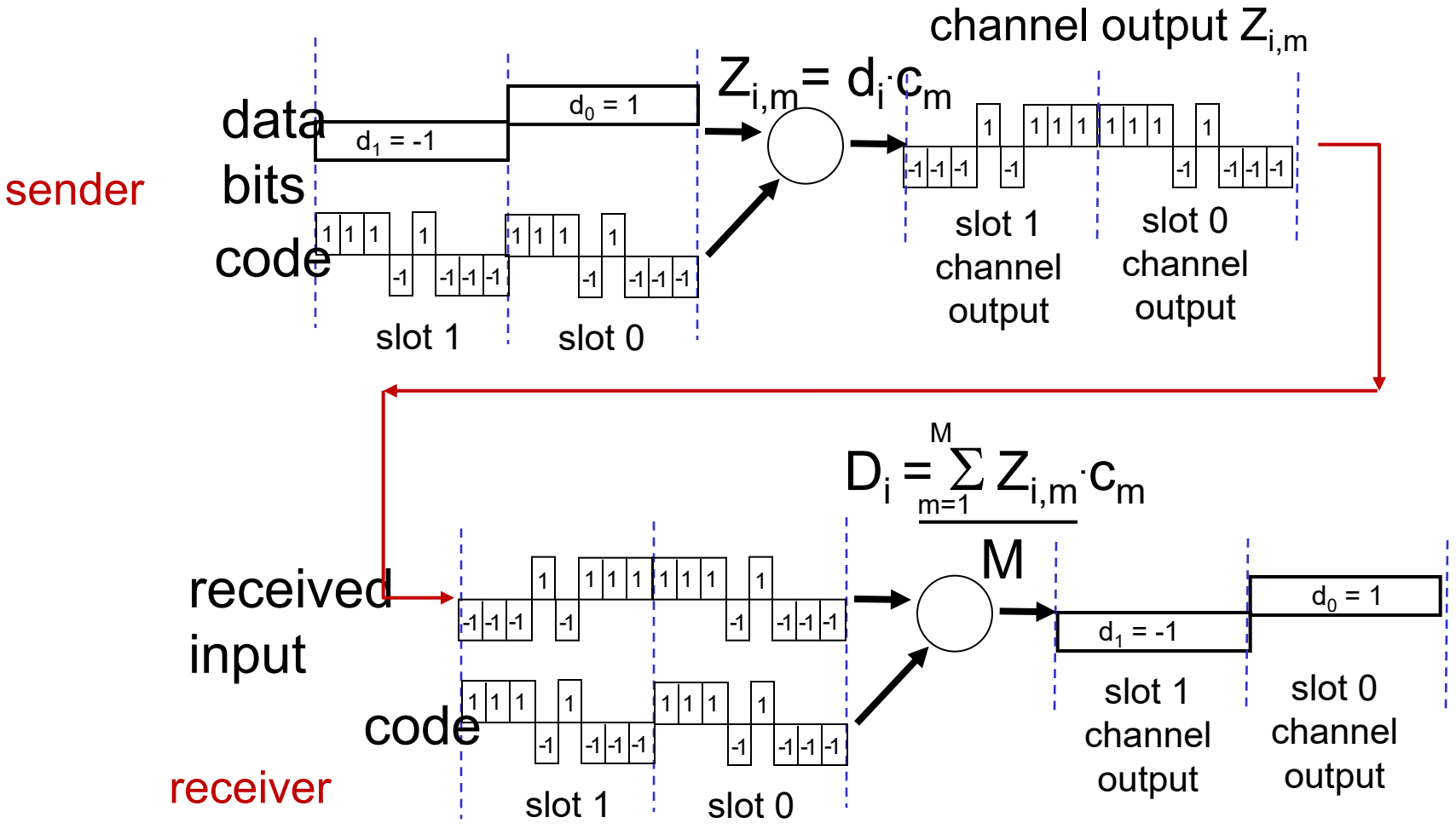
# *Multiplexing : TDMA



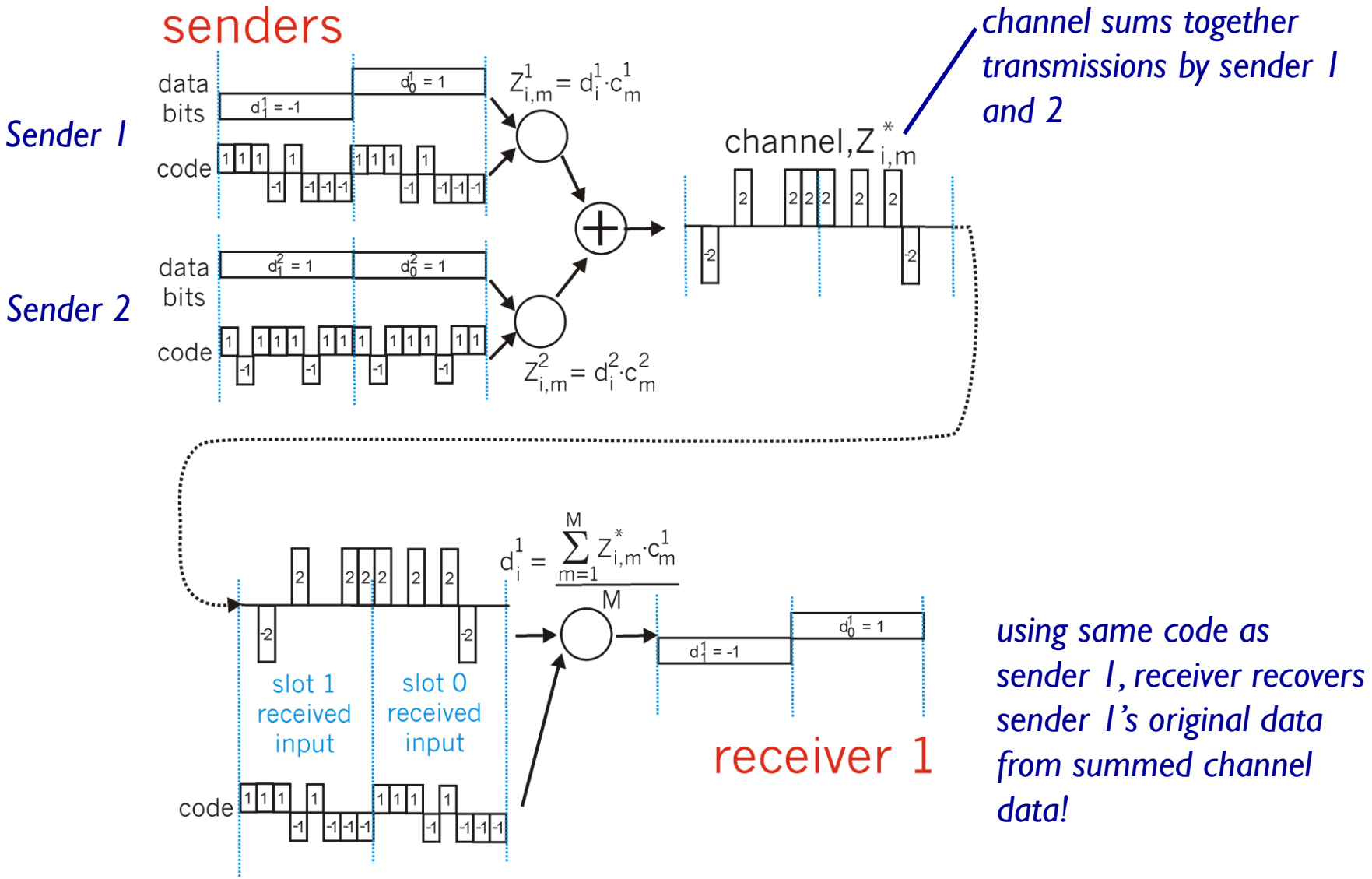
# *Multiplexing : Code Division Multiple Access (CDMA)

- unique "code" assigned to each user; i.e., code set partitioning
  - all users share same frequency, but each user has own "chipping" sequence (i.e., code) to encode data
  - allows multiple users to "coexist" and transmit simultaneously with minimal interference (if codes are "orthogonal")
- *encoded signal* = (original data) X (chipping sequence)
- *decoding:* inner-product of encoded signal and chipping sequence

# *Multiplexing : CDMA encode/decode



# *Multiplexing : CDMA: two-sender interference



# *Comparison of commonly used media

| Transmission medium | Transfer method | Rate/ Band | Transmission distance | Performance | Price | Application |
|---|---|---|---|---|---|---|
| Twisted pair | Broadband Baseband | ≤1Gb/s | Anolog: 10km digtal: 500m | Better | Low | Analog / digital signal transmission |
| 50Ω Coaxial cable | Baseband | 10Mb/s | <3km | Better | Lower | Baseband digital signal |
| 75Ω Coaxial cable | Broadband | ≤450MHz | 100km | Better | Lower | Analog TV, data and audio |
| Optical fiber | Baseband | 40Gb/s | 20km以上 | Very good | Higher | Long - distance high - speed data transmission |
| Microwave | Broadband | 4-6GHz | Several hundred km | Good | Medium | Remote communication |
| Satellite | Broadband | 1-10GHz | 18000 km | Very good | High | Remote communication |

# *Commonly used physical layer interface standards

- EIA-232-E interface standard
- RS-449 interface standard
- RS-485 interface standard
- CAN interface standard
- PROFIBUS interface standard
- …….

# Recap of the Physical Layer

- The function of the physical layer and the issues to be considered
- Four important characteristics of the physical layer
- The contents of the physical layer protocol
- Some Basic Concepts in Data Communication
- Nyquist's Law and Shannon's Formula
- Signal Coding Technology and Multiplexing Technology
- Commonly used transmission medias
- Commonly used physical layer interface standards

# Exercise-1

1.In the physical layer interface characteristics, the time sequence used to describe the completion of each function is

A. Mechanical Characteristics

B. Electrical Characteristics

C. Functional Characteristics

D. Procedural Characteristics

2. Among the following options, that do not belong to the definition scope of physical layer interface specification is

A. interface shape  B. pin function

C. physical address  D. signal level

# Exercise-2

1.Under the condition of no noise, if the frequency bandwidth of a communication link is 3kHz and QAM modulation technology with four phases and four amplitudes in each phase is adopted, the maximum data transmission rate of the communication link is

A. 12kbps B. 24kbps C. 48kbps D. 96kbps

2. If the data transmission rate of a communication link is 2400bps and 4-phase modulation is adopted, the baud rate of the link is

A. 600 baud B. 1200 baud

C. 4800 baud D. 9600 baud

# Exercise-2

3. If the frequency bandwidth connecting R2 and R3 link is 8kHz and the signal-to-noise ratio is 30dB, and the actual data transmission rate of the link is about 50% of the maximum data transmission rate, the actual data transmission rate of the link is about

A. 8kbps B. 20kbps C. 40kbps D. 80kbps

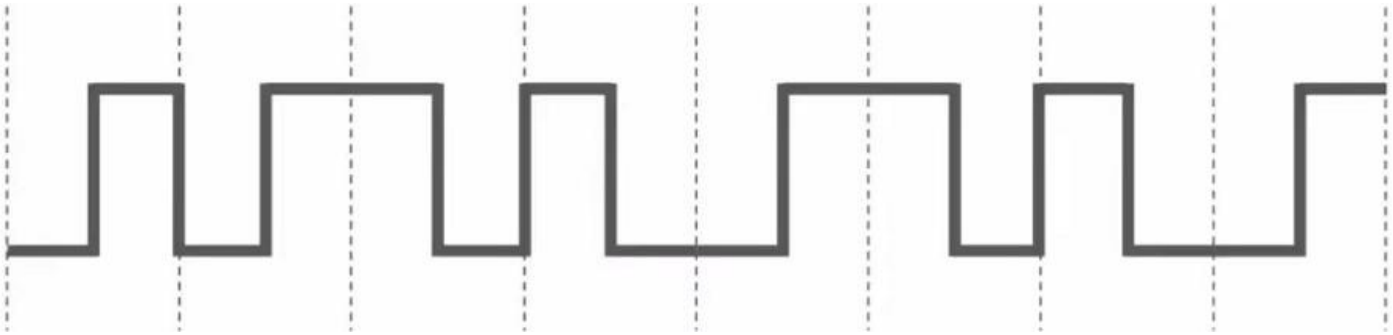
2. Among the following factors, which will not affect the channel data transmission rate is

A. signal to noise ratio B. frequency bandwidth

C. symbol speed D. signal propagation speed

# Exercise-3

1.If the following figure shows the signal waveform received by the 10Base-T network interface card, the bit string received by the network interface card is



A. 00110110 B. 10101101 C. 01010010 D. 11000101